

Worksheet 1
MTH 203 (Multivariate Calculus)
Date: 18-08-2026

Q1. Let a be any positive real number. Using the principle of mathematical induction prove that $(1+a)^n \geq 1+na$ for $n \in \mathbb{N}$. Moreover, using this inequality prove that for a positive real number c the sequence $(c^{1/n}) \rightarrow 1$.

Q2. Let (x_n) be a sequence of real number. Is it true that if the sequence $(|x_n|)$ converges, then the sequence (x_n) converges? If yes prove it or provide the counterexample.

Q3. Determine whether the following sequences are convergent or not:

(a) $x_n = \frac{n}{n+1}$;

(b) $x_n = \frac{(-1)^n n}{n+1}$

(c) $x_n = \sin n$

Q4. Determine whether the following series are convergent or not with short explanation:

(a) $\sum \frac{1}{n^p}$ where $p < 1$;

(b) $\sum \frac{1}{n^p}$ where $p \geq 2$;

(c) $\sum (1 + \frac{1}{n^2})$;

(d) $\sum -ne^{-n}$.

Q5. Suppose (x_n) be a convergent sequence. Prove that for every $\epsilon > 0$ there exists a positive integer $N(\epsilon) \in \mathbb{N}$ such that $|x_n - x_m| < \epsilon$ for all $n, m \geq N(\epsilon)$.

Q6. Let $(\mathbb{R}^n, ||\cdot||)$ be a vector space with the Euclidean norm. Show that for $x, y \in \mathbb{R}^n$ we have

$$\frac{1}{2}(|x+y||^2 + (||x-y||)^2) = ||x||^2 + ||y||^2.$$

Q7. In each of the following cases, let S be the all points (x, y) in the plane satisfying the given inequalities. Make a sketch showing the set S .

(a) $|x| < 1$ and $|y| < 1$;

(b) $(x^2 + y^2 - 1)(4 - x^2 - y^2) > 0$;

(c) $y > x^2$ and $|x| < 2$.

Q8. Prove that

$$\lim_{n \rightarrow \infty} \left(\frac{1}{(n+1)^2} + \frac{1}{(n+2)^2} + \cdots + \frac{1}{(2n)^2} \right) = 0.$$

Q9. Which of the following series converges or diverges? Give reasons for your answers.

(a) $\sum \frac{\ln(n+1)}{n+1};$

(b) $\sum \frac{\sec^{-1}(n)}{n^{1.3}};$

(c) $\sum \frac{3^{n-1}+1}{3^n}.$

Q10. Which of the series converge absolutely, which converge, and which diverge? Give reasons for your answers.

(a) $\sum \frac{(-1)^n}{1+\sqrt{n}};$

(b) $\sum (-1)^n (\sqrt{n+1} - \sqrt{n});$

(c) $\sum (-1)^n \frac{(n+1)^n}{(2n)^n};$

(d) $\sum (-1)^n \frac{(2n)!}{2^n n! n}.$